

Ehsan Sharafian

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Professional Summary

Mechanical Engineering PhD candidate specializing in intelligent wearable systems for human movement analysis and digital health. Experienced in building end-to-end solutions that integrate IMU-based sensing, embedded hardware, smartphone applications, machine learning, and haptic feedback for gait analysis and rehabilitation. Skilled in Python, MATLAB, Android Studio/Java, and PyTorch, with six peer-reviewed journal publications.

Education

PhD	University of Maine , Mechanical Engineering - GPA: 4.00/4.00 - Dissertation: Smart Real-Time Gait Training System	Orono, ME Expected Dec 2027
MS	Amirkabir University of Technology (Tehran Polytechnic) , Mechanical Engineering - GPA: 3.65/4.00 - Thesis: Trajectory optimization of a robot arm in joint and Cartesian spaces	Feb 2020
BS	University of Tehran , Mechanical Engineering	Jul 2017

Technical Skills

Programming & Data Analysis: Python, Java, C/C++, NumPy, Pandas, scikit-learn, PyTorch, SPSS

Mobile & Software: MATLAB, Simulink, TwinCAT, Android Studio, Unity, Git

Wearable & Embedded Systems: IMU sensors, Arduino IDE, SparkFun modules, vibrotactile motors, PCB modules

Biomechanics & Signal Processing: Gait analysis, human activity recognition, sensor fusion, feature extraction, OpenSim

Robotics: Surgical robotics, parallel manipulators, inverse kinematics, dynamics, control systems

CAD: SolidWorks

Research & Engineering Experience

Graduate Research Assistant , Biorobotics and Biomechanics Lab, University of Maine - Led the development of real-time wearable sensing systems and Android-based mobile applications for gait analysis, activity recognition, fall-risk assessment, and home-based rehabilitation using IMUs, haptic feedback, and smartphone-deployed feedback platforms. - Built a smartphone-deployed gait assessment system using a single foot-mounted IMU to reconstruct foot-height trajectories, achieving 96.08% locomotion classification accuracy, less than 1.1 cm cumulative height error, and 112 ms latency. - Engineered a lightweight human activity recognition system using two IMUs, biomechanical feature extraction, and supervised classifiers including SVM and Naive Bayes, achieving 99.1% accuracy for normal locomotion and 93.1% accuracy for atypical stair-negotiation strategies in older adults. - Architected a home-based haptic feedback gait-training system integrating IMUs, vibrotactile motors, custom Android applications, Arduino/SparkFun modules, and PCB circuitry for adults aged 65+. - Demonstrated up to 30% improvement in stride length and walking speed during a single-session haptic feedback study, with minimal cognitive demand. - Led IRB-approved human-subject validation studies with 85+ participants across gait analysis, activity recognition, and haptic feedback protocols. - Developed real-time algorithms and data pipelines in Android Studio/Java, Python, and MATLAB, combining signal processing, sensor fusion, biomechanical modeling, and machine learning for mobile health applications.	Orono, ME Sept 2023 – present
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Mechanical and Control Systems Engineer, Sina Robotics and Medical Innovators Co.

Tehran, Iran
Dec 2022 – Sept 2023

- Developed PLC-based control logic for a telesurgical laparoscopic robotic platform using Beckhoff TwinCAT and IEC 61131-3 Structured Text, with MATLAB/Simulink validation of inverse kinematics and initialization workflows.
- Built a Python/Kivy graphical user interface on Raspberry Pi for surgeon-facing robot control, system guidance, warnings, and improved human-robot interaction during testing and demonstrations.
- Coordinated hands-on validation sessions with **6 experienced surgeons**, using training, troubleshooting, and usability feedback to refine surgeon-facing robotic system workflows.

Laboratory Research Assistant, New Technology Research Center, Amirkabir University of Technology

Tehran, Iran
Nov 2020 – Nov 2022

- Designed and assembled robotics and mechatronics setups for student research projects, integrating hardware, CAD, MATLAB workflows, and technical documentation.
- Mentored **6 undergraduate students** across **5 research teams**, guiding equipment use, troubleshooting, documentation, and project execution.
- Coordinated technical preparation for the **9th International Conference on Robotics and Mechatronics**, including lab demonstrations, equipment readiness, and presentation logistics.

Selected Honors

Flagship Fellowship Award, University of Maine

Sep 2024 – Sep 2025

Graduate Student Government Award for conference proposal

Mar 2025

Leadership & Service

Graduate Senator, Department of Mechanical Engineering, University of Maine

Sept 2025 – present

President, Iranian Graduate Student Association, University of Maine

Sept 2024 – Apr 2026

Publications

- **6 peer-reviewed journal articles** (3 first-author) in wearable sensing, biomechanics, rehabilitation engineering, and robotics.
- **Sensors** (2026) – Real-time foot-height estimation and activity classification
- **IEEE Access** (2025) – Lightweight IMU-based activity recognition for older adults
- **IEEE TNSRE** (2025) – Haptic feedback effects on gait and cognitive load
- **Journal of Biomechanics** (2025) – Thigh extension and leg coordination
- **International Journal of Robotics and Automation** (2023) – Trajectory optimization of a robot arm
- **Robotica** (2022) – Screw theory-based constraint wrench analysis